Shopify Clone: Technical Documentation

Project Overview and Architecture

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Empowering Merchants with a Premium Shopify Clone

The project aims to provide a comprehensive technical solution that enables merchants to create, manage, and scale online stores with ease.

Merchant Empowerment

Tools to build and manage digital storefronts from scratch.

Scalability

Designed to handle growth from initial store setup to high-volume sales.

Premium Experience

Focus on delivering a high-end, responsive shopping experience.

Comprehensive Multi-Tenant Architectural Scope

Multi-Tenancy

Support for multiple isolated merchant stores on a single platform with shared infrastructure.

Full-Stack Coverage

Includes front-end storefronts, merchant admin dashboards, and complex back-end logic.

Deployment Ready

Detailed installation and configuration guides for seamless setup by system administrators.

Targeted Insights for Developers and Stakeholders

Developers

Deep dive into the codebase and multi-tenant architecture for future expansion and customization.

System Administrators

Clear instructions for environment setup, deployment, and ongoing system maintenance.

Stakeholders

High-level overview of system capabilities, market-readiness, and business value proposition.

Multi-Tenant Architecture via Subdomain Routing

Dynamic Routing

Next.js Middleware intercepts requests to resolve store identity based on the host header.

```
store1.localhost:3000
↓
/store/store1/home
```

System Layers

Platform Layer

Central hub for user acquisition, landing pages, and global platform administration.

Admin Layer

Secure merchant dashboard for managing products, orders, and store-specific settings.

Storefront Layer

High-performance, dynamic customer-facing pages optimized for the shopping experience.

Cutting-Edge Technology Stack for Modern Web Apps

Frameworks

Next.js 16 and React 19 for industry-leading performance and server-side capabilities.

Styling

Tailwind CSS 4 with Framer Motion for smooth, glassmorphic animations and modern UI.

Data Management

PostgreSQL (Supabase) with Prisma ORM for robust, type-safe data modeling.

Authentication

Next-Auth for secure, multi-provider user sign-in and session management.

Seamless Third-Party Service Integration

stripe

Stripe Payments

Powering secure checkout flows and real-time payment webhooks for global transactions.

Cloudinary Media

Managing high-performance image storage and automatic optimization for product catalogs.

Supabase Foundation

Providing a robust, scalable PostgreSQL foundation for all relational data and store settings.

Simplified Installation and System Requirements

Prerequisites

- Node.js 18.17 or later
- pnpm (recommended) or npm
- Supabase (PostgreSQL) Account
- Cloudinary Account (Media)
- Stripe Account (Payments)

System Specs

Memory:
4GB RAM min (8GB recommended)

Disk Space:
~500MB for app and dependencies

OS Support:
Windows, macOS, or Linux

Rapid Deployment in Four Easy Steps

01 Clone & Install

Get the source code and install dependencies.

```
git clone <repo-url>
pnpm install
```

02 Database Setup

Initialize Prisma and push to database.

```
npx prisma generate
npx prisma db push
```

03 Environment Config

Configure API keys in your .env file.

```
cp .env.example .env
# Edit variables
```

04 Launch

Start the server and access the platform.

```
npm run dev
```

Intuitive UI Across Platform, Admin, and Storefront

Platform Home

A futuristic landing page designed with high-impact visuals to convert visitors into active merchants.

Merchant Dashboard

A dark-themed, glassmorphic interface providing data-driven tools for efficient store management.

Customer Storefront

A minimalist, clean design focused entirely on product presentation and seamless checkout.

Robust Core Functionality and Advanced Analytics

Store Creation

Instant initialization of new stores with custom subdomains and automated setup.

Product Management

Full variant support, inventory tracking, and optimized image handling via Cloudinary.

Order Fulfillment

Real-time tracking and management of customer purchases from checkout to delivery.

Advanced Analytics

Visualized sales data and customer insights to help merchants make informed decisions.

Conclusion: A Scalable Future for E-commerce

Ready to Scale

Architecture built for growth, supporting multiple tenants with isolated environments and shared efficiency.

Modern Stack

Leveraging the best of the web ecosystem with Next.js, React, and Prisma for a production-ready foundation.

Extensible

Clear documentation and modular design provide a solid base for future feature development and customization.

THANK YOU FOR YOUR ATTENTION